PHISHPULSE CAMPAIGN-NOVELTY-GUIDED CONFORMAL PREDICTION AND SELECTIVE RISK CONTROL FOR ZERO-DAY PHISHING URL DETECTION

A PROJECT REPORT (PHASE-I)

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in partial fulfillment of the requirement

for the award of the degree

of

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING

K.S.RANGASAMY COLLEGE OF TECHNOLOGY

(Autonomous)

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NOVEMBER - 2026

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BONAFIDE CERTIFICATE

Certified that this Project Report (Phase-I) titled **"PHISHPULSE CAMPAIGN-NOVELTY-GUIDED CONFORMAL PREDICTION AND SELECTIVE RISK CONTROL FOR ZERO-DAY PHISHING URL DETECTION"** is the bonafide work of **REVANTH P (2303737710421041)** and **SABARISH M(2303737710421043)** who carried out the project under my supervision. Certified further, that to the best of my knowledge the work reported herein does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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We jointly declare that the Project Report (Phase-I) on **"PHISHPULSE CAMPAIGN-NOVELTY-GUIDED CONFORMAL PREDICTION AND SELECTIVE RISK CONTROL FOR ZERO-DAY PHISHING URL DETECTION"** is the result of original work done by us and best of our knowledge, similar work has not been submitted to **"ANNA UNIVERSITY, CHENNAI"** for the requirement of Degree of Bachelor of Engineering. This project report is submitted on the partial fulfilment of the requirement of the award of Degree of Bachelor of Engineering.

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ACKNOWLEDGEMENT

We wish to express our sincere gratitude to our honourable Chairman **Thiru. R. SRINIVASAN, B.B.M.,** for providing immense facilities at our institution.

We would like to express our sincere gratitude to our Vice Chairman, **Thiru. K. S. SACHIN, MBA (UK).,** for providing a supportive environment for the completion of our project work.

We are truly grateful to express our gratitude to our beloved Principal **Dr. R. GOPALAKRISHNAN, M.E., Ph.D.,** for the encouragement given towards the progress and completion of our project.

We proudly render our immense gratitude to the Head of the Department **Dr. K. POONGODI, M.E., Ph.D.,** for her effective leadership, encouragement and guidance in the project.

We are highly indebted to provide our heartfelt thanks to our supervisor **Mrs. M. SARADHA, M.E.,** Assistant Professor for her valuable idea, supportive guidance and encouragement throughout the project.

We wish to extend our sincere thanks to all faculty members of **COMPUTER SCIENCE AND ENGINEERING** for their valuable suggestions, kind co-operation and constant encouragement for successful completion of this project.

We wish to acknowledge the help received from various Departments, individuals, and the AICTE IDEA Lab facilities in the successful completion of this project.

ABSTRACT

Phishing is a major cybersecurity threat in which attackers create deceptive URLs to trick users into visiting malicious websites and revealing sensitive information such as usernames, passwords, and financial details. Traditional phishing detection methods may depend on blacklists or complex detection systems, which can have limitations when identifying newly created or previously unseen phishing URLs. Therefore, an efficient, lightweight, and interpretable phishing URL detection framework is required.

The proposed project, PhishPulse, aims to detect phishing URLs using a combination of lexical and structural URL features without fetching webpage content or performing external website scraping. The system extracts a 21-dimensional hybrid feature vector from the input URL, including URL syntax, character entropy, and domain structure-related characteristics.To improve interpretability, the framework incorporates SHAP-based explainability techniques to identify the features that influence each prediction. The system also provides readable risk indicators and a user-friendly dashboard to display the classification result and its supporting explanations. By combining lightweight machine learning, hybrid feature extraction, and explainable predictions, PhishPulse aims to provide an efficient and understandable approach to phishing URL detection.

Keywords: Phishing Detection, URL Classification, Machine Learning, Hybrid Features, Explainable AI, SHAP, Cybersecurity.

This project supports **SDG 9: Industry, Innovation and Infrastructure** by using lightweight machine learning and explainable AI to detect phishing URLs and strengthen cybersecurity. It also contributes to **SDG 16: Peace, Justice and Strong Institutions** by helping users identify potentially malicious URLs, reduce the risk of online fraud, and promote a safer digital environment.

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LIST OF SYMBOLS AND ABBREVIATIONS

SYMBOLS

Symbol	Unit	Description
U	URL length	Characters
N	Number of URL features	Count
H	Character entropy	Bits per character
P	Prediction probability	Dimensionless (0–1)
T	Prediction time	seconds (s)
A	Accuracy	Percentage (%)
Pr	Precision	Percentage (%)
R	Recall	Percentage (%)

ABBREVIATIONS

Abbreviation	Expansion
AI	Artificial Intelligence
ML	Machine Learning
URL	Uniform Resource Locator
SHAP	SHapley Additive exPlanations
RF	Random Forest
DT	Decision Tree
LR	Logistic Regression
TN	True Negative
TP	True Positive
FN	False Negative
FP	False Positive
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure

CHAPTER 1

INTRODUCTION

Phishing URLs can trick users into visiting fraudulent websites and exposing sensitive information. Therefore, this project, "PhishPulse Campaign-Novelty-Guided Conformal Prediction and Selective Risk Control for Zero-Day Phishing URL Detection" aims to identify potentially malicious URLs using machine learning algorithms. The proposed system extracts a 21-dimensional hybrid feature vector based on lexical and structural URL characteristics, including URL syntax, suspicious tokens, character entropy, and domain structure. The system uses Random Forest, Decision Tree, and Logistic Regression algorithms to classify URLs as phishing or legitimate. SHAP-based explainability is incorporated to identify the features influencing the predictions and present relevant risk indicators through a user-friendly dashboard. By combining lightweight URL analysis, machine learning classification, and explainable predictions, PhishPulse aims to improve phishing awareness and support safer online browsing..

1.1 BACKGROUND

Phishing is a common cybersecurity threat in which attackers create fraudulent URLs to deceive users into visiting malicious websites and sharing sensitive information such as usernames, passwords, and financial details. These URLs are often designed to resemble legitimate website addresses, making them difficult to identify through manual inspection.

Traditional phishing detection techniques frequently rely on blacklists containing previously identified malicious URLs. However, these methods may not detect newly created or previously unseen phishing URLs. Machine learning provides an alternative approach by analyzing URL characteristics and identifying patterns associated with phishing activities.

Machine learning provides an effective approach to phishing URL detection by learning patterns from labelled datasets containing phishing and legitimate URLs. Different URL characteristics can be extracted and represented as numerical features for classification.

Lexical features describe the URL's textual properties, such as its length, special characters, and suspicious tokens. Structural features represent characteristics related to the arrangement of URL components and domain structure. Combining these characteristics can provide useful information for distinguishing potentially malicious URLs from legitimate ones.

The proposed project, PhishPulse Campaign-Novelty-Guided Conformal Prediction and Selective Risk Control for Zero-Day Phishing URL Detection, focuses on detecting phishing URLs using lexical and structural features. The system extracts a 21-dimensional hybrid feature vector from the input URL without fetching webpage content or performing external website scraping.

To improve transparency, the framework incorporates SHAP-based explainability techniques to identify the features that influence predictions. The system presents the classification results and relevant risk indicators through a user-friendly dashboard.

1.2 PROBLEM STATEMENT

Phishing is a serious cybersecurity threat that involves creating deceptive URLs to trick users into visiting fraudulent websites and revealing sensitive information, including usernames, passwords, banking credentials, and personal details. As online services continue to grow, phishing attacks have become increasingly difficult to identify because attackers manipulate URL structures, use misleading domain names, and include suspicious words or characters to make malicious links appear legitimate

Traditional phishing detection techniques often rely on blacklists containing previously identified malicious URLs. Although these methods can identify known threats, they may fail to recognize newly created phishing URLs that have not yet been reported. Furthermore, some advanced detection approaches depend on webpage content, external network requests, or computationally intensive models, which can increase processing requirements and implementation complexity. Another major limitation is the lack of interpretability in certain machine learning systems, where users receive a prediction without understanding the reasons behind the classification.

The primary problem addressed by this project is the need for a lightweight, efficient, and interpretable phishing URL detection framework that can analyze URL-based characteristics, compare machine learning models, and provide understandable explanations for its predictions. The system aims to improve the transparency of phishing detection and support users in making more informed decisions when encountering suspicious URLs.

1.3 NEED FOR THE PROJECT

A phishing URL detection system is useful because it can help users identify suspicious links before visiting potentially malicious websites. Early identification may help users avoid revealing sensitive information and reduce the risk of online fraud.

The need for this project can be explained through the following points:

1. Automatic Phishing URL Detection

The system analyses URL characteristics and automatically classifies URLs as phishing or legitimate. This reduces the need for manual URL inspection and supports faster identification of potential threats.

2. Detecting Suspicious URL Patterns

Phishing URLs may contain suspicious keywords, unusual characters, and misleading domain names. Analysing these patterns helps identify characteristics commonly associated with phishing attempts.

3. Reducing Dependence on Webpage Content

The system analyses URL strings directly without fetching webpage content or performing HTML scraping. This reduces dependence on external webpage resources during URL analysis.

4. Identifying Previously Unseen Threats

Blacklist-based methods may not recognize newly created phishing URLs. Feature-based machine learning analysis may help identify suspicious URLs that have not previously been reported.

5. Combining Lexical and Structural Features

The system extracts a 21-dimensional hybrid feature vector containing lexical and structural URL characteristics. Combining these features provides useful information for distinguishing phishing URLs from legitimate URLs.

6. Comparing Machine Learning Algorithms

The project uses Random Forest, Decision Tree, and Logistic Regression for URL classification. Comparing their performance helps evaluate their suitability for phishing detection.

7. Explaining Prediction Results

A classification result alone may not explain why a URL is considered suspicious. SHAP-based explanations identify important contributing features and improve the transparency of model predictions.

8. Displaying Risk Indicators

The dashboard presents the predicted URL classification and relevant risk indicators. This helps users understand suspicious URL characteristics through a clear and accessible interface.

9. Evaluating System Performance

The system is evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. Prediction time is also measured to assess the efficiency of URL analysis.

10. Promoting Safer Online Browsing

Phishing attacks can cause financial loss, identity theft, and unauthorized access to personal accounts. The system helps users make more informed decisions by identifying potentially malicious URLs and presenting understandable explanations.

The project is needed to explore how these features can work together in a single system. Its actual usefulness and performance will be determined through implementation and testing.

1.4 OBJECTIVES

The main objective of this project is to develop an interpretable and lightweight phishing URL detection system that analyses URL characteristics using machine learning algorithms and provides understandable explanations for its predictions.

The specific objectives are:

- To develop a phishing URL detection framework using Random Forest, Decision Tree, and Logistic Regression algorithms.
- To extract a 21-dimensional hybrid feature vector from URLs using lexical and structural characteristics.
- To analyse URL properties such as URL length, suspicious tokens, special characters, character entropy, and domain structure.
- To classify input URLs as phishing or legitimate based on the extracted features.
- To perform URL analysis without fetching webpage content or extracting HTML information.
- To incorporate SHAP-based explainability techniques to identify the features influencing model predictions

- To compare the performance of the selected machine learning algorithms using appropriate evaluation metrics.
- To measure system performance using accuracy, precision, recall, F1-score, ROC-AUC, and prediction time.
- To improve prediction transparency and help users make informed decisions when encountering potentially malicious URLs.

These objectives describe the planned work. Their completion will depend on the implementation and testing of the system.

1.5 SCOPE OF THE PROJECT

The scope of this project is to develop an interpretable and lightweight phishing URL detection system that identifies potentially malicious URLs by analysing their lexical and structural characteristics.

The proposed system will extract a 21-dimensional hybrid feature vector from the input URL and use machine learning algorithms to classify URLs as phishing or legitimate. The framework will also incorporate SHAP-based explainability to identify the features influencing the classification results.

The project will cover the following areas:

1. URL Feature Extraction

The system will extract lexical and structural features directly from the input URL. These features will include URL length, suspicious tokens, special characters, character entropy, and domain structure.

2. Hybrid Feature Construction

The extracted URL characteristics will be combined into a 21-dimensional hybrid feature vector. This representation will provide the input required by the machine learning models for URL classification.

3. Machine Learning Classification

The project will implement Random Forest, Decision Tree, and Logistic Regression algorithms to classify URLs as phishing or legitimate. Their performance will be compared using suitable evaluation metrics.

4. URL-Based Analysis

The system will analyse the URL string without fetching webpage content or performing HTML scraping. This limits the analysis to URL-based characteristics rather than webpage content.

5. Explainable AI Integration

SHAP-based explainability techniques will be used to identify the features that influence model predictions. The explanations will help users understand the reasons behind a URL's classification.

6. Risk Indicator Presentation

The system will display the predicted classification and relevant risk indicators through a user-friendly dashboard. This will help users interpret the analysis results more easily.

7. Model Performance Evaluation

The system will be evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. Prediction time will also be measured to assess the efficiency of URL analysis.

8. System Testing and Validation

The framework will be tested using suitable phishing and legitimate URL datasets. Testing will examine classification performance, prediction consistency, explanation quality, and the system's ability to analyse input URLs.

Its main purpose is to analyse URL characteristics, classify potentially malicious links, and provide understandable explanations to support informed cybersecurity decisions.

CHAPTER 2

LITERATURE REVIEW

2.1 OBSERVATION OF EXISTING RESEARCH

Existing research on phishing URL detection has focused on machine learning algorithms, deep learning techniques, URL-based feature extraction, and explainable artificial intelligence. Several studies have explored lexical and structural URL characteristics to distinguish phishing URLs from legitimate ones. Machine learning models have been used to improve classification performance, while explainability techniques have been investigated to understand the factors influencing model predictions. However, challenges remain in detecting newly created phishing URLs, reducing computational complexity, and providing understandable explanations for classification results. Some approaches depend on webpage content or complex models, which may increase processing requirements. Therefore, the proposed PhishPulse project aims to combine a 21-dimensional hybrid feature vector with lightweight machine learning algorithms and SHAP-based explanations. The system will classify URLs without fetching webpage content and provide risk indicators through a user-friendly dashboard. Its effectiveness will be evaluated using classification metrics and prediction time to assess detection performance, efficiency, and interpretability.

2.1.1 Explainable AI for Phishing URL Detection: A Bayesian-Optimized Stacking Ensemble Framework with SHAP-Guided Feature Learning for URL Detection

"Explainable AI for Phishing URL Detection: A Bayesian-Optimized Stacking Ensemble Framework with SHAP-Guided Feature Learning for URL Detection" (2026) focuses on identifying potentially malicious URLs using machine learning and explainable artificial intelligence. Phishing URLs may contain misleading domain names, suspicious keywords, unusual characters, and complex structures that make them difficult to distinguish from legitimate URLs. Combining multiple machine learning models can help improve classification, while explainability can provide information about the factors influencing predictions.

Ensemble learning and feature engineering can be used to combine the predictions of multiple models and identify important URL characteristics. Bayesian optimization can support the selection of suitable model parameters, while SHAP can help explain individual predictions by showing the contribution of different features

This research topic is relevant to the proposed PhishPulse project because it highlights the importance of combining phishing detection with explainable predictions. The proposed system will use a 21-dimensional hybrid feature vector containing lexical and structural URL characteristics and apply Random Forest, Decision Tree, and Logistic Regression for classification. SHAP-based explanations will help identify the features influencing predictions, while a dashboard will display classification results and risk indicators. The system's performance will be assessed using suitable evaluation metrics and prediction time.

2.1.2 Phishing Webpage Detection Using Structured URL Generation

"Phishing Webpage Detection Using Structured URL Generation" (2026) focuses on detecting phishing-related threats by examining URL structures and their patterns. Phishing attackers may construct URLs that imitate legitimate websites by manipulating domain names, adding misleading words, and using complex paths or subdomains. Analysing the structure of URLs can provide useful information for identifying potentially malicious links and reducing the risk of users accessing fraudulent websites.

Structured URL analysis can support the identification of suspicious patterns, including unusual domain arrangements, excessive URL components, and misleading combinations of characters. Machine learning techniques can use these characteristics to distinguish potentially malicious URLs from legitimate ones. However, variations in URL construction and similarities between phishing and legitimate links can make accurate classification challenging, especially when encountering newly generated URLs.

This research topic is relevant to the proposed PhishPulse project because it emphasizes the value of URL structure analysis in phishing detection. The proposed system will Random Forest, Decision Tree, and Logistic Regression for classification. SHAP-based explanations will help identify the features influencing predictions, while a dashboard will display classification results and risk indicators. The system will analyse URLs without fetching webpage content or performing HTML scraping, and its performance will be assessed using suitable evaluation metrics and prediction time.

2.1.3 Phishing URL Detection and Interpretability With Machine Learning: A Cross-Dataset Approach

"Phishing URL Detection and Interpretability With Machine Learning: A Cross-Dataset Approach" (2026) focuses on detecting phishing URLs while examining the interpretability and generalization of machine learning models. Phishing URLs frequently use deceptive domain names, suspicious keywords, and unusual structural patterns to appear trustworthy.

Machine learning algorithms can classify URLs using extracted lexical and structural features, while explainability techniques can help users understand the reasons behind individual predictions. Cross-dataset evaluation can provide insights into how well a trained model performs on data that differs from its training dataset. However, differences in dataset composition, feature distributions, and phishing patterns can reduce model performance and make generalization difficult.

This research topic is relevant to the proposed PhishPulse project because it highlights the importance of both model interpretability and reliable evaluation. The proposed system will extract a 21-dimensional hybrid feature vector. SHAP-based explanations will identify the features influencing predictions, and the dashboard will display classification results and risk indicators. The system's performance will be assessed using suitable evaluation metrics, with appropriate testing used to examine its ability to classify URLs beyond the training data.

2.1.4 Adversarial-Resilient Lightweight Phishing URL Detection

"Adversarial-Resilient Lightweight Phishing URL Detection" (2026) focuses on developing phishing URL detection methods that remain useful when attackers deliberately modify URL patterns to avoid identification. Phishing attackers may change domain names, insert additional characters, use misleading words, or modify URL structures to make malicious links appear legitimate. A lightweight detection approach is important for supporting efficient URL analysis without requiring excessive computational resources.

Machine learning models can identify suspicious URL characteristics, while adversarial testing can help evaluate how changes to URL patterns affect classification. Lightweight feature extraction and efficient algorithms can reduce processing requirements and support faster predictions. However, attackers may continuously change their URL construction techniques, and small modifications to a URL can affect model predictions. Therefore, detection performance must be evaluated carefully against different URL patterns.

This research topic is relevant to the proposed PhishPulse project because it emphasizes lightweight processing and the challenges of detecting changing phishing patterns. The proposed system will construct a 21-dimensional hybrid feature vector from lexical and structural URL characteristics and use Random Forest, Decision Tree, and Logistic Regression for classification.

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2.1.5 Across the Spectrum: In-Depth Review AI-Based Models for Phishing Detection

"Across the Spectrum: In-Depth Review AI-Based Models for Phishing Detection" (2025) focuses on examining artificial intelligence and machine learning approaches used to identify phishing threats. Phishing attacks can target users through deceptive URLs, fraudulent websites, and misleading online communications. Reviewing different detection techniques helps researchers understand how automated methods are developed, what features they use, and which challenges affect their practical application.

AI-based phishing detection may involve traditional machine learning, ensemble learning, deep learning, and feature-based classification techniques. These approaches can analyse URL characteristics, webpage information, or other indicators to identify potentially malicious activity. However, differences in datasets, evaluation methods, computational requirements, and interpretability can make it difficult to compare the effectiveness of different approaches directly. Such challenges highlight the importance of selecting appropriate algorithms and evaluation metrics for a specific application.

This research topic is relevant to the proposed PhishPulse project because it provides a broader perspective on the available approaches to phishing detection. The proposed system will focus on URL-based analysis using a 21-dimensional hybrid feature vector and compare Random Forest, Decision Tree, and Logistic Regression. SHAP-based explanations will help make predictions understandable, while a dashboard will display classification results and risk indicators.

2.1.6 Machine Learning for Early Detection of Phishing URLs in Parked Domains: An Approach Applied to a Financial Institution

"Machine Learning for Early Detection of Phishing URLs in Parked Domains: An Approach Applied to a Financial Institution" (2025) focuses on identifying potentially malicious URLs associated with parked domains, particularly in a financial security context. Parked domains may remain inactive or display limited content, while some can be misused in suspicious online activities.

Machine learning techniques can be applied to URL characteristics and domain-related indicators to distinguish potentially suspicious links from legitimate ones. Early detection can help security teams identify risks before users interact with harmful resources. However, parked domains may provide limited information for analysis, and legitimate inactive domains can share characteristics with suspicious domains.

This research topic is relevant to the proposed PhishPulse project because it highlights the value of early phishing identification using URL-related information.

The proposed system will extract a 21-dimensional hybrid feature vector based on lexical and structural URL characteristics and use Random Forest, Decision Tree, and Logistic Regression for classification.requirements.

2.1.7 Evaluating the Impact of Feature Engineering in Phishing URL Detection: A Comparative Study of URL, HTML, and Derived Features

"Evaluating the Impact of Feature Engineering in Phishing URL Detection: A Comparative Study of URL, HTML, and Derived Features" (2025) focuses on understanding how different types of features influence phishing detection performance. Phishing websites can imitate legitimate services through deceptive URL structures and webpage content. Selecting appropriate features is important because the quality and relevance of the input data can affect a machine learning model's ability to distinguish phishing URLs from legitimate ones.

A Comparing these feature groups can help determine which characteristics contribute to classification. However, HTML-based analysis may require webpage retrieval and additional processing, increasing system complexity.

This research is relevant to PhishPulse because it highlights the importance of feature selection and representation. The proposed system will combine lexical and structural URL characteristics into a 21-dimensional hybrid feature vector without fetching webpage content or performing HTML scraping.

2.1.8 Explainable AI for URL Threat Detection in Healthcare Cybersecurity

"Explainable AI for URL Threat Detection in Healthcare Cybersecurity" (2025) focuses on applying explainable artificial intelligence to identify potentially harmful URLs in healthcare-related digital environments. Healthcare organizations use online platforms to manage patient information, appointments, and other sensitive services, making them potential targets for phishing attacks.

Machine learning models can examine URL characteristics to identify suspicious patterns, while explainability techniques can help users understand the factors influencing classification results. Such explanations may improve transparency and assist cybersecurity teams in reviewing potentially malicious links.

This research is relevant to PhishPulse because it emphasizes transparent phishing detection. The proposed system will use SHAP-based explanations to identify influential URL features and present classification results and risk indicators through a dashboard.

2.1.9 An Explainable Feature Selection Framework for Web Phishing Detection with Machine Learning

"*An Explainable Feature Selection Framework for Web Phishing Detection with Machine Learning"* (2025) focuses on selecting relevant features for phishing detection while making the model's decisions easier to understand. Phishing websites and URLs often contain suspicious words, misleading domain names, and unusual structural characteristics. However, using too many irrelevant or redundant features may increase processing requirements and complicate the classification process.

Explainability techniques can further show how selected features influence the final classification, helping users understand the reasons for a prediction. However, the usefulness of selected features may depend on the dataset and the characteristics of the phishing attacks being studied.

This research is relevant to PhishPulse because the proposed system also relies on feature engineering and explainability. It will use a 21-dimensional hybrid URL feature vector and SHAP-based explanations to make classification results easier to interpret. The system will be evaluated using suitable performance metrics and prediction time to assess whether its selected URL features support effective and interpretable phishing detection.

2.1.10 Detection of Phishing Websites and Emails Using Explainable Machine Learning Models

"Detection of Phishing Websites and Emails Using Explainable Machine Learning Models" (2025) focuses on identifying phishing threats across websites and email communications using machine learning. Phishing attacks commonly use deceptive links, misleading messages, and imitations of trusted organizations to persuade users to disclose sensitive information. Detecting these threats automatically can help reduce the risks of credential theft, financial fraud, and unauthorized access.

The Explainability techniques can identify the features that contribute to a model's prediction and help users understand the classification. However, website, email, and URL analysis may require different input features and processing methods.

This research is relevant to PhishPulse because it highlights the importance of interpretable phishing detection. The proposed system concentrates specifically on URL-based classification and uses SHAP-based explanations to identify features associated with suspicious links.

2.1.11 Lightweight and Explainable Phishing URL Detection Using Feature Selection and XGBoost

"Lightweight and Explainable Phishing URL Detection Using Feature Selection and XGBoost" (2025) focuses on detecting phishing URLs while considering computational efficiency and model interpretability. Phishing URLs may contain suspicious tokens, misleading domain names, excessive special characters, and unusual structures. A lightweight detection framework can help analyse such URLs efficiently, while feature selection can reduce unnecessary input information and support a more manageable classification process.

The feature selection techniques can identify relevant URL characteristics, and XGBoost can be used to classify URLs based on the selected features. Explainability methods can help identify which features contribute to individual predictions. However, the effectiveness of a model depends on the dataset, feature quality, and evaluation procedure.

This research is relevant to PhishPulse because it combines efficient classification with understandable results. The proposed system will compare Random Forest, Decision Tree, and Logistic Regression and use SHAP-based explanations to support transparent URL analysis.

2.1.12 Phishing URL Detection with Neural Networks: An Empirical Study

"Phishing URL Detection with Neural Networks: An Empirical Study" (2024) focuses on examining the use of neural network models to identify potentially malicious URLs. Phishing URLs may contain misleading domain names, suspicious tokens, unusual character sequences, and complex structures that are difficult to identify through manual inspection. Neural networks can learn patterns from training data and use them to distinguish phishing URLs from legitimate ones, making them a possible approach for automated cybersecurity applications.

Empirical evaluation can help determine how well the model performs on previously labelled URLs. However, neural networks may require careful training, suitable parameter selection, and sufficient data. Their computational requirements and interpretability can also be concerns when developing a lightweight system that must explain why a URL has been classified as suspicious.

This research is relevant to PhishPulse because it highlights the importance of model evaluation. The proposed system will use Random Forest, Decision Tree, and Logistic Regression to compare classification performance and provide explanations for their predictions.

2.1.13 Phishing Website Detection Using Deep Learning Models

"Phishing Website Detection Using Deep Learning Models" (2024) focuses on applying deep learning techniques to identify phishing websites that imitate legitimate online services. Phishing websites may use misleading domain names, copied layouts, and deceptive links to convince users that a fraudulent website is trustworthy. Automated detection can help identify potential threats and reduce the likelihood of users exposing sensitive information.

These models can learn relationships between input characteristics and known phishing labels, allowing them to classify previously unseen examples when the learned patterns generalize well. However, deep learning methods may require substantial training data and computational resources. Their decision-making processes can also be difficult to interpret, making clear explanations important for cybersecurity applications.

This research is relevant to PhishPulse because it provides an alternative approach to phishing detection. The proposed system focuses on lightweight machine learning and direct URL analysis without fetching webpage content.

2.1.14 Enhancing Online Safety: Phishing URL Detection Using Machine Learning and Explainable AI

"Enhancing Online Safety: Phishing URL Detection Using Machine Learning and Explainable AI" (2024) focuses on identifying potentially malicious URLs through automated classification and understandable predictions. Phishing attacks often involve links that imitate legitimate websites, use suspicious keywords, or contain unusual domain structures. Users may have difficulty distinguishing these URLs from trustworthy links, especially when attackers carefully design deceptive addresses.

Machine learning algorithms can learn from labelled URL data and classify input URLs as phishing or legitimate. Explainable AI techniques can identify the characteristics that influence the model's decision, making predictions easier to interpret. However, detection systems may encounter previously unseen phishing URLs, misleading legitimate URLs, and changes in attack strategies.

This research is relevant to PhishPulse because it combines automated detection with interpretability. The proposed system will extract hybrid URL features, compare three machine learning algorithms, and display classifications and risk indicators through a dashboard.

2.1.15 Phishing URL Detection Using Machine Learning and Deep Learning

"Phishing URL Detection Using Machine Learning and Deep Learning" (2024) focuses on applying data-driven algorithms to distinguish potentially malicious URLs from legitimate web addresses. Phishing URLs often use misleading domain names, suspicious words, unusual symbols, and complicated structures to deceive users. Automated detection can help identify such characteristics more consistently than manual inspection alone.

Machine learning models often rely on engineered features, whereas deep learning models may learn representations from suitable input data. Their performance and computational requirements depend on the chosen algorithms, dataset, and implementation.

This research is relevant to PhishPulse because it highlights the importance of comparing detection methods. The proposed system will focus on three conventional machine learning algorithms and use SHAP-based explanations to improve prediction transparency.

2.1.16 Explainable Machine Learning for Phishing Detection

"*Explainable Machine Learning for Phishing Detection"* (2024) focuses on making phishing detection systems more transparent by explaining how machine learning models arrive at their predictions. Phishing attacks can use deceptive URLs and fraudulent online resources to steal passwords, banking information, and other sensitive data. Although machine learning can automate the identification of suspicious patterns, users may find it difficult to trust a classification when the system provides only a phishing or legitimate label without explaining the decision.

Explainable machine learning techniques can identify influential features and provide additional insight into the classification process. The quality of these explanations depends on the model, input representation, and explanation method.

This research is relevant to PhishPulse because explainability is a central component of the proposed framework. SHAP-based explanations will help identify the URL characteristics influencing predictions and present relevant risk indicators to users.

2.1.17 Detection of Malicious URLs Using Machine Learning

"Detection of Malicious URLs Using Machine Learning" (2024) focuses on identifying URLs that may lead to harmful online resources through automated classification. Malicious URLs can be used for phishing, malware distribution, fraudulent activities, and other cybersecurity threats. These URLs may contain suspicious keywords, unusual character combinations, misleading domains, or complex structures.

Feature extraction and classification models can help distinguish potentially harmful links from legitimate URLs. However, a model designed for general malicious URL detection may require different training data and evaluation procedures from a phishing-specific model.

This research is relevant to PhishPulse because phishing detection is a specific application of URL classification. The proposed system will focus on phishing URLs and analyse lexical and structural characteristics using a 21-dimensional feature vector.

2.1.18 Phishing URL Detection Using Machine Learning

"*Phishing URL Detection Using Machine Learning*" (2023) focuses on automatically identifying suspicious URLs that may direct users to fraudulent websites. Phishing attackers often create links that imitate legitimate services, manipulate domain names, and include suspicious words or unusual characters. These deceptive URLs can make it difficult for users to recognize potential threats through visual inspection alone.

Features such as URL length, special characters, suspicious tokens, and domain structure can help models distinguish phishing URLs from legitimate ones. However, phishing patterns change over time, and some malicious URLs closely resemble legitimate links.

This research is relevant to PhishPulse because it provides the foundation for its classification pipeline. The proposed system extends URL-based machine learning by combining lexical and structural features with SHAP-based explanations and a user-friendly dashboard.

2.1.19 Phishing or Not Phishing? A Survey on the Detection of Phishing Websites

"Phishing or Not Phishing? A Survey on the Detection of Phishing Websites" (2023) focuses on reviewing methods used to distinguish phishing websites from legitimate websites. Phishing remains a significant cybersecurity concern because attackers can imitate trusted organizations and persuade users to disclose sensitive information.

Different detection methods have been developed to identify fraudulent websites, including blacklist-based techniques, URL analysis, machine learning, and approaches that examine webpage content.

Reviewing these methods helps identify challenges such as evolving attack techniques, false-positive classifications, dataset limitations, and computational complexity. The usefulness of each approach depends on its intended application and evaluation conditions.

This research is relevant to PhishPulse because it provides background for selecting suitable detection techniques. The proposed system focuses on direct URL analysis, machine learning classification, and explainability to support more transparent phishing detection.

2.1.20 On Phishing URLs Detection Using Feature Extension

"On Phishing URLs Detection Using Feature Extension" (2023) focuses on improving phishing URL detection by extending the features extracted from URLs. The researchers use the TextRank algorithm to build a feature extension library and incorporate additional information into URL representations. The approach combines a BERT layer with a convolutional neural network (CNN) to learn both comprehensive and local patterns in URLs.

The proposed method aims to improve classification accuracy and stability while reducing overfitting. However, its use of BERT and CNN introduces additional model complexity compared with conventional machine learning algorithms, which may make it less suitable for applications requiring lightweight processing.

This research is relevant to PhishPulse because it demonstrates the importance of feature representation in phishing URL detection. The proposed PhishPulse system uses a 21-dimensional hybrid feature vector containing lexical and structural URL characteristics and compares Random Forest, Decision Tree, and Logistic Regression. Unlike the feature-extension approach, PhishPulse focuses on lightweight classification and uses SHAP-based explanations to make predictions easier to understand.

2.1.21 Phishing URL Detection: A Real-Case Scenario Through Login URLs

"*Phishing URL Detection: A Real-Case Scenario Through Login URLs"* (2022) focuses on identifying potentially malicious URLs associated with login pages and authentication-related resources. Login URLs are common targets for phishing attacks because attackers may create deceptive links that imitate banking services, educational portals, social media platforms, and other trusted websites.

URL analysis can help identify suspicious domain names, misleading paths, and unusual character patterns. However, legitimate login URLs may also contain similar characteristics, so multiple features must be considered during classification.

This research is relevant to PhishPulse because the proposed system analyses URL structure and suspicious tokens. Combining these features within a hybrid feature vector may help distinguish potentially malicious links from legitimate login URLs.

2.1.22 Phishing Websites Detection Using Machine Learning with URL Analysis

"Phishing Websites Detection Using Machine Learning with URL Analysis" (2022) focuses on identifying phishing websites by analysing the URLs associated with them. Attackers often construct deceptive web addresses that contain misleading domain names, suspicious words, unusual character sequences, or multiple subdomains. These characteristics can help distinguish potentially malicious URLs from legitimate links.

Feature-based classification can automate the detection process, but performance may be affected by similarities between legitimate and phishing URLs. Reliable evaluation is necessary to determine how well the model handles different URL patterns.

This research is relevant to PhishPulse because it uses URL analysis as the basis for classification. The proposed system combines lexical and structural features and provides SHAP-based explanations to help users understand its predictions.

2.1.23 A Deep Learning-Based Framework for Phishing

"A Deep Learning-Based Framework for Phishing" (2022) focuses on applying deep learning techniques to identify phishing-related threats through automated analysis. Phishing attacks often use deceptive links, fraudulent websites, and misleading online content to trick users into revealing sensitive information. Since these attacks can change their appearance and structure, automated detection methods are important for identifying suspicious patterns.

Deep learning methods can learn complex patterns from suitable datasets, but their effectiveness depends on the input data, model architecture, and experimental evaluation. The specific techniques and findings must be confirmed using the complete publication.

This research may be relevant to PhishPulse if the original paper addresses URL or website phishing detection. The proposed system focuses on a lightweight URL-based framework using a 21-dimensional hybrid feature vector and explainable machine learning.

2.1.24 An Effective Detection Approach for Phishing Websites Using URL and HTML Features

"An Effective Detection Approach for Phishing Websites Using URL and HTML Features" (2022) focuses on identifying phishing websites by combining information from URLs and webpage HTML content. Phishing attackers may use misleading domain names and suspicious URL structures while designing webpages that imitate trusted services. Analysing both the URL and HTML can provide different types of information about a website.

Combining these feature types may provide more information for classification. However, HTML-based analysis generally requires webpage retrieval and additional processing, which can increase computational requirements and dependence on external resources.

This research is relevant to PhishPulse because it demonstrates an alternative feature-based approach. The proposed system analyses URL strings directly and uses lexical and structural features without fetching webpage content or performing HTML scraping.

2.1.25 URL Phishing Detection Using Machine Learning Techniques Based on URLs Lexical Analysis

"URL Phishing Detection Using Machine Learning Techniques Based on URLs Lexical Analysis" (2021) focuses on identifying phishing URLs by examining the textual characteristics of web addresses. Phishing attackers may include misleading words, unusual character combinations, excessive symbols, and suspicious patterns to make fraudulent URLs appear legitimate. Lexical analysis provides a way to examine these characteristics directly from a URL string.

These characteristics can be converted into numerical features and supplied to classification algorithms. However, lexical analysis alone may not capture all domain relationships or structural patterns associated with phishing attacks.

This research is relevant to PhishPulse because lexical analysis forms an important part of its feature extraction process. The proposed system extends this approach by combining lexical and structural characteristics into a 21-dimensional hybrid feature vector and incorporating SHAP-based explanations.

2.2 COMPARATIVE ANALYSIS OF EXISTING METHODS

The reviewed research covers several approaches to phishing URL detection, including lexical analysis, structural feature extraction, traditional machine learning, deep learning, ensemble learning, feature selection, and explainable artificial intelligence. Each approach addresses a different aspect of identifying potentially malicious URLs and distinguishing them from legitimate web addresses.

Lexical and structural feature-based techniques can identify suspicious URL patterns by analysing URL length, special characters, suspicious tokens, character entropy, and domain structure. Traditional machine learning methods, such as Random Forest, Decision Tree, and Logistic Regression, can use these features to classify URLs. Deep learning and ensemble-based approaches offer alternative methods for learning complex patterns, although their computational requirements and implementation complexity may vary. Feature selection can help identify relevant characteristics, while cross-dataset evaluation can provide insights into model generalization.

Some approaches focus on improving classification performance, while others emphasize lightweight processing, adversarial resilience, or prediction interpretability. Explainable AI techniques, such as SHAP, can help identify the features influencing model predictions. URL-based approaches can also reduce dependence on webpage content and HTML extraction. The proposed PhishPulse project combines a 21-dimensional hybrid feature vector, three machine learning algorithms, and SHAP-based explanations to provide phishing URL classification with understandable risk indicators. Its performance will be evaluated using suitable classification metrics and prediction time.

The following table provides a high-level comparison of the research approaches related to phishing URL detection. It compares their main purposes and relevance to the proposed PhishPulse project. The comparison is based on the paper titles and topics currently available; the exact methods and reported performance should be verified from the original papers before the final report is submitted.

Research approach	Main purpose	Relevance to the proposed project
Lexical analysis	Identify suspicious patterns in URL text, including length, keywords, and special characters.	Supports the extraction of lexical features from input URLs.
Structural feature extraction	Examine domain structure, subdomains, and URL components.	Contributes structural characteristics to the 21-dimensional hybrid feature vector.
Traditional machine learning	Classify URLs using algorithms trained on labelled data.	Supports comparison of Random Forest, Decision Tree, and Logistic Regression.
Ensemble learning	Combine predictions from multiple models to support classification.	Provides a research direction for comparing model performance.
Feature selection	Identify relevant features and reduce redundant information.	Helps importance of URL characteristics for classification.
Explainable artificial intelligence	Explain the factors influencing machine learning predictions.	Supports SHAP-based explanations
Cross-dataset evaluation	Examine model performance across different datasets.	rovides a direction for evaluating model generalization
Lightweight detection	Reduce processing while maintaining useful classification performance.	Aligns with PhishPulse's focus on efficient URL

Table 1.1 Comparative Analysis

The comparison suggests that an effective phishing URL detection system involves more than selecting a machine learning model. It also requires appropriate URL feature extraction, efficient classification, understandable explanations, and systematic performance evaluation. The proposed PhishPulse project aims to integrate these functions into a single workflow to identify potentially malicious URLs and support safer online browsing.

Overall Literature Review

S. No.	Paper Name	Year	Research Gap	Research Gap Overcome in PhishPulse
1	Explainable AI for Phishing URL Detection: A Bayesian-Optimized Stacking Ensemble Framework with SHAP-Guided Feature Learning for URL Detection	2026	Advanced stacking ensembles and Bayesian optimization increase implementation complexity.	Uses campaign-novelty-guided conformal prediction and selective risk control for zero-day phishing detection.
2	Phishing Webpage Detection Using Structured URL Generation	2026	URL generation and completion add processing complexity.	Analyzes original URLs and uses novelty-aware prediction without URL generation.
3	Phishing URL Detection and Interpretability With Machine Learning: A Cross-Dataset Approach	2026	Cross-dataset performance alone does not ensure reliable zero-day detection.	Uses conformal prediction and selective classification to manage uncertain predictions.
4	Adversarial-Resilient Lightweight Phishing URL Detection	2026	Models may fail when attackers manipulate URL structures to evade detection.	Uses campaign novelty to identify potentially unfamiliar phishing patterns and control prediction risk.
5	Across the Spectrum: In-Depth Review AI-Based Models for Phishing Detection	2025	A review identifies research trends but does not implement a complete detection system.	Integrates novelty assessment, conformal prediction, and selective risk control into a detection framework.
6	Machine Learning for Early Detection of Phishing URLs in Parked Domains: An Approach Applied to a Financial Institution	2025	Early detection relies on domain-monitoring data and specialized external sources.	Evaluates submitted URLs using campaign-novelty information to support zero-day detection.
7	Evaluating the Impact of Feature Engineering in Phishing URL Detection: A Comparative Study of URL, HTML, and Derived Features	2025	HTML-based features require webpage retrieval and additional processing.	Focuses on URL-based analysis and uncertainty-aware predictions without requiring webpage HTML.
8	Explainable AI for URL Threat Detection in Healthcare Cybersecurity	2025	The research focuses on healthcare cybersecurity rather than general-purpose URL detection.	Targets zero-day phishing detection using campaign novelty and selective prediction.
9	An Explainable Feature Selection Framework for Web Phishing Detection with Machine Learning	2025	Feature selection alone does not ensure reliable predictions for previously unseen campaigns.	Uses conformal prediction to quantify prediction uncertainty and supports abstention on uncertain cases.
10	Detection of Phishing Websites and Emails Using Explainable Machine Learning Models	2025	Combining website and email analysis increases input and processing complexity.	Focuses on URL-based phishing detection with campaign novelty and controlled prediction decisions.
11	Lightweight and Explainable Phishing URL Detection Using Feature Selection and XGBoost	2025	The focus on XGBoost leaves scope for better handling of novel threats and uncertain predictions.	Adds novelty-guided conformal prediction and selective risk control for unfamiliar phishing

12	Phishing URL Detection with Neural Networks: An Empirical Study	2024	Neural network models may be more complex than necessary for lightweight detection.	Emphasizes uncertainty-aware decisions for zero-day URLs rather than relying only on classification.
13	Phishing Website Detection Using Deep Learning Models	2024	Deep learning can require more computing resources and may struggle with unseen threats.	Uses campaign novelty and selective prediction to manage unfamiliar URLs.
14	Enhancing Online Safety: Phishing URL Detection Using Machine Learning and Explainable AI	2024	High accuracy does not guarantee reliable predictions for new phishing campaigns.	Incorporates novelty assessment, conformal prediction, and abstention for uncertain cases.
15	Phishing URL Detection Using Machine Learning and Deep Learning	2024	Models may generalize poorly to previously unseen phishing URLs.	Focuses on zero-day detection and prediction-risk control.
16	Explainable Machine Learning for Phishing Detection	2024	Complex models may be difficult for users to understand and may produce uncertain predictions.	Provides a framework for handling prediction uncertainty through selective classification.
17	Detection of Malicious URLs Using Machine Learning	2024	Classification may not provide reliable decisions for previously unseen phishing campaigns.	Uses campaign-novelty-guided conformal prediction to manage unfamiliar URLs.
18	Phishing URL Detection Using Machine Learning	2023	Traditional methods may fail to identify new phishing URLs.	Uses campaign novelty and selective risk control to address zero-day threats.
19	Phishing or Not Phishing? A Survey on the Detection of Phishing Websites	2023	Existing techniques vary in their ability to detect new threats and manage uncertainty.	Combines novelty assessment, conformal prediction, and selective classification.
20	A Deep Learning-Based Framework for Phishing Website Detection	2022	Multiple detection mechanisms increase system complexity and may not handle unseen campaigns reliably.	Provides a focused framework for zero-day detection with uncertainty-aware decisions.
21	Phishing URL Detection: A Real-Case Scenario Through Login URLs	2022	Models trained on limited URL types may misclassify unfamiliar login URLs.	Uses novelty-guided prediction to identify potentially unfamiliar URL patterns.
22	Phishing Websites Detection Using Machine Learning with URL Analysis	2022	Limited dataset diversity may reduce generalization to new phishing campaigns.	Evaluates campaign novelty and uses conformal prediction to manage uncertainty.
23	A Deep Learning-Based Framework for Phishing	2022	Multiple detection mechanisms increase system complexity and may not handle unseen campaigns reliably.	Provides a focused framework for zero-day detection with uncertainty-aware decisions.
24	An Effective Detection Approach for Phishing Websites Using URL and HTML Features	2022	Some methods do not account for uncertainty when encountering unseen phishing URLs.	Uses selective prediction to avoid making unsupported high-confidence decisions.
25	URL Phishing Detection Using Machine Learning Techniques	2021	Results on one dataset may not generalize to newer phishing URLs.	Focuses on zero-day phishing detection through campaign novelty.

Table 1.2 Literature Review

2.3 RESEARCH GAP

The reviewed paper topics show that phishing URL detection research covers lexical analysis, structural feature extraction, machine learning classification, deep learning, explainable AI, feature selection, computational efficiency, and cross-dataset evaluation. These areas provide useful foundations for developing an interpretable and lightweight phishing URL detection system.

However, the following points will need to be examined more closely in the original papers to establish a research gap supported by evidence.

1. Combining lexical and structural URL features

Phishing detection can examine textual patterns and URL structure to identify suspicious links. The proposed PhishPulse project will combine lexical and structural characteristics into a 21-dimensional hybrid feature vector to investigate their usefulness for classification.

2. Identifying previously unseen phishing URLs

phishing URLs may not appear in existing datasets or blacklists. The proposed project will investigate how machine learning models classify URLs based on their extracted features, while evaluating their performance on appropriate test data.

3. Balancing classification performance and computational efficiency

Complex detection approaches may require additional computational resources. The proposed system will compare Random Forest, Decision Tree, and Logistic Regression to assess their classification performance and prediction time.

4. Improving the interpretability of predictions

Some machine learning models provide classifications without clearly explaining the factors influencing their decisions. The proposed project will incorporate SHAP-based explanations to identify influential URL features and present understandable risk indicators.

5. Reducing dependence on webpage content

Some phishing detection approaches analyse HTML content or other webpage information, which may require additional processing and network requests. PhishPulse will analyse URL characteristics directly without fetching webpage content or performing HTML scraping.

6. Evaluating detection reliability

A phishing detection system may incorrectly classify legitimate URLs or fail to identify malicious ones. The proposed project will evaluate its models using accuracy, precision, recall, F1-score, ROC-AUC, and prediction time to examine classification performance and processing efficiency.

These are potential research-gap directions based on the reviewed topics and the proposed system design. They should be confirmed against the original papers before being presented as established gaps in existing research.

2.4 CONCLUSION FROM THE LITERATURE SURVEY

This chapter has outlined different research directions related to phishing URL detection. The reviewed topics include lexical analysis, structural feature extraction, traditional machine learning, deep learning, ensemble learning, feature selection, explainable AI, and cross-dataset evaluation. Each approach offers a different way to identify suspicious URL patterns, classify potentially malicious links, or improve the interpretability of detection results.

The literature provides a foundation for selecting a suitable detection approach for the proposed PhishPulse project. Random Forest, Decision Tree, and Logistic Regression are being considered because the system aims to classify URLs using a 21-dimensional hybrid feature vector containing lexical and structural characteristics. SHAP-based explainability will be incorporated to identify the features influencing predictions. However, the effectiveness of the selected models must be established through implementation and experimental evaluation.

The proposed project aims to combine URL preprocessing, hybrid feature extraction, machine learning classification, SHAP-based explanations, and dashboard presentation in a single workflow. The system is designed to analyse URLs without fetching webpage content or performing HTML scraping. These functions are planned features, and their classification performance, interpretability, and prediction time have not yet been verified.

The next chapter discusses the system analysis, including the problem definition, existing approaches, their limitations, and the hardware and software requirements for developing the proposed PhishPulse system.

CHAPTER 3

PROPOSED METHODOLOGY

The proposed methodology aims to develop an interpretable and lightweight phishing URL detection system that identifies potentially malicious URLs and helps users understand the associated risks. The system uses lexical and structural URL characteristics to construct a 21-dimensional hybrid feature vector. The extracted features are processed by Random Forest, Decision Tree, and Logistic Regression algorithms to classify URLs as phishing or legitimate. SHAP-based explainability is incorporated to identify the features influencing the predictions and provide understandable risk indicators. The system analyses URLs directly without fetching webpage content or performing HTML scraping. A user-friendly dashboard displays the classification results and explanations. The overall methodology includes dataset collection, URL preprocessing, hybrid feature extraction, machine learning model training, phishing URL classification, explainability analysis, dashboard development, and system evaluation.

3.1 OVERVIEW OF THE PROPOSED APPROACH

The proposed system aims to identify potentially malicious phishing URLs and provide understandable explanations to help users make informed decisions when encountering suspicious links. The system uses a machine learning-based approach to analyse URL characteristics and classify URLs as phishing or legitimate. The extracted features and classification results are presented through a user-friendly dashboard, with SHAP-based explanations highlighting the factors influencing each prediction.

The proposed approach consists of dataset collection, URL preprocessing, hybrid feature extraction, model training, phishing URL classification, explainability analysis, and dashboard development. The system combines lexical and structural URL characteristics into a 21-dimensional hybrid feature vector and uses Random Forest, Decision Tree, and Logistic Regression for classification. The approach is designed to provide lightweight URL analysis without fetching webpage content or performing HTML scraping. Model performance and prediction time will be evaluated experimentally.

The overall workflow is:

1. URL dataset collection and preparation.
2. URL preprocessing.
3. SHAP-based explanation and risk indicator generation..
4. Phishing URL classification using Random Forest, Decision Tree, and Logistic Regression.
5. Dashboard presentation and system performance evaluation.

3.2 REQUIREMENTS ANALYSIS

The The proposed PhishPulse system requires both hardware and software components to analyse URLs, classify potential phishing threats, and display understandable prediction results.

Hardware requirements:

- Computer or laptop for model training and system execution.
- Sufficient RAM and storage to process the URL dataset and save trained models.
- Processor capable of performing URL feature extraction and machine learning inference.
- Internet connection for accessing datasets and using the web-based dashboard.
- Optional GPU for faster model training on larger datasets.

Software requirements:

- Python for feature extraction, model development, and backend processing.
- Pandas and NumPy for dataset handling and numerical operations.
- Scikit-learn for Random Forest, Decision Tree, and Logistic Regression.
- SHAP for explaining the features influencing model predictions.
- HTML, CSS, and JavaScript for developing the user interface.
- Matplotlib or Plotly for displaying charts and risk indicators.

The system should support URL input, extract a 21-dimensional hybrid feature vector, classify URLs as phishing or legitimate, and display prediction results with relevant risk indicators. SHAP-based explanations should help users understand the features influencing each prediction.

3.3 DATA COLLECTION

A labelled dataset containing phishing and legitimate URLs is required to develop and evaluate the PhishPulse detection system. The dataset should include a variety of URL patterns, domain structures, suspicious tokens, and character combinations to help the models learn the differences between phishing and legitimate URLs.

Each URL is assigned a corresponding class label indicating whether it is phishing or legitimate. The dataset is then divided into training, validation, and testing subsets. Duplicate URLs and unsuitable records should be handled during preprocessing to improve data quality and reduce the risk of biased evaluation.

The training set is used to train the Random Forest, Decision Tree, and Logistic Regression models. The validation set can be used to select model settings and classification thresholds, while the test set is used to evaluate performance on previously unseen URLs. Data splitting should be performed carefully to prevent information leakage between the subsets.

The collected URLs are processed to extract a 21-dimensional hybrid feature vector containing lexical and structural characteristics. These features are used for classification without fetching webpage content or extracting HTML. The dataset supports model comparison using accuracy, precision, recall, F1-score, ROC-AUC, and prediction time.

3.4 DATA / INPUT PROCESSING

The The collected phishing and legitimate URLs are checked for missing values, duplicate records, invalid entries, and incorrect class labels. The dataset is cleaned and prepared for feature extraction.

The dataset is divided into training, validation, and testing subsets to reduce data leakage and support reliable model evaluation. Any preprocessing operations that learn parameters from the data should be fitted using the training set and then applied to the validation and testing sets as appropriate.

During prediction, the user-provided URL is validated and processed to extract a 21-dimensional hybrid feature vector. The extracted features include URL length, suspicious tokens, special characters, character entropy, and domain structure. The feature vector is passed to the trained Random Forest, Decision Tree, or Logistic Regression model to classify the URL as phishing or legitimate. The model's output is then used to generate the prediction result and corresponding SHAP-based explanation.

The final results are displayed through the dashboard, including the predicted class and relevant risk indicators. Since the system analyses the URL directly without fetching webpage content or extracting HTML, the processing workflow focuses on URL-based features. The same feature extraction procedure must be applied consistently during training and prediction to ensure reliable results.

3.5 METHODOLOGY

The methodology combines phishing URL feature extraction, machine learning classification, and explainability analysis. The system analyses the characteristics of an input URL, classifies it as phishing or legitimate, and presents the factors influencing the prediction through a user-friendly dashboard.

3.5.1 Phishing URL Classification Using Machine Learning

Random Forest, Decision Tree, and Logistic Regression are used as the baseline machine learning algorithms for phishing URL detection. The labelled dataset is preprocessed, and a 21-dimensional hybrid feature vector is extracted from the lexical and structural characteristics of each URL. The models are trained using the prepared training data and evaluated using separate test data.

For each input URL, the system extracts the required features and passes them to the trained model. The model predicts whether the URL is potentially phishing or legitimate. The classification result and, where supported, the prediction probability are used to present the outcome on the dashboard.

3.5.2 Hybrid URL Feature Extraction

After After receiving a URL, the system extracts lexical and structural characteristics to construct a 21-dimensional hybrid feature vector. Lexical features examine the URL's text, including its length, suspicious tokens, and special characters. Structural features examine URL components, such as domain structure and subdomain patterns. Character entropy can also be used to identify unusual character distributions.

. The same feature extraction procedure must be applied consistently during model training and prediction to maintain compatibility between the input data and the trained models.

The system analyses the URL string directly without fetching webpage content or extracting HTML. This reduces dependence on external webpage requests and keeps the detection process focused on URL-based information.

3.5.3 Explainability Analysis and Risk Indicator Generation

The explainability module uses SHAP (SHapley Additive exPlanations) to identify the features that influence the machine learning model's predictions. For tree-based models, TreeExplainer can be used where supported, while a suitable SHAP explainer can be selected for Logistic Regression.

After classification, the system analyses the feature contributions to help explain why a URL was classified as potentially phishing or legitimate. Relevant indicators, such as suspicious tokens, unusual character patterns, and domain structure, can be displayed alongside the prediction. The explanations depend on the trained model and the extracted features; they should not be treated as definitive proof that a URL is malicious or safe.

The dashboard presents the predicted class, relevant risk indicators, and available explanation results in an understandable format. This stage improves the transparency of the detection process and helps users make more informed decisions when encountering suspicious links.

3.6 TOOLS AND TECHNOLOGIES

The The PhishPulse project uses the following technologies:

- Python: Model development, URL feature extraction, data preprocessing, and backend logic.
- Scikit-learn: Training and evaluation of Random Forest, Decision Tree, and Logistic Regression models.
- Pandas and NumPy: Dataset processing, data cleaning, and numerical feature handling.
- SHAP: Explaining the features influencing machine learning predictions.
- Flask or FastAPI: Backend API for receiving URLs and returning classification results.
- HTML, CSS, and JavaScript: Frontend structure, styling, and user interaction.
- Matplotlib or Plotly: Visualisation of model performance and URL risk indicators.
- CSV: Storage and management of labelled URL datasets and evaluation results.
- JSON: Communication of URL inputs, prediction results, and explanation data between the frontend and backend.

3.7 EVALUATION APPROACH

The proposed system is evaluated at both the model and application levels.

The proposed PhishPulse system is evaluated at both the model and application levels to assess its classification performance, prediction speed, and explainability.

At the model level, accuracy, precision, recall, F1-score, and ROC-AUC are used to measure phishing URL classification performance. Prediction time is measured to assess the processing speed of the system. These metrics help evaluate how effectively the models identify phishing URLs while minimising false-positive and false-negative predictions.

At the application level, the system is tested using phishing and legitimate URLs that were not used during model training. The evaluation considers correct classifications, false-positive predictions, missed phishing URLs, and the consistency of the results displayed on the dashboard. Random Forest, Decision Tree, and Logistic Regression are compared to identify differences in their classification performance.

The results are used to identify model limitations and assess whether PhishPulse provides useful, efficient, and interpretable phishing URL detection. All performance values, including accuracy, precision, recall, F1-score, ROC-AUC, and prediction time, should be reported from actual experiments rather than assumed results.

CHAPTER 4

SYSTEM DESIGN

The system design explains how the different components of the PhishPulse: An Interpretable and Lightweight Framework for Phishing URL Detection Using URL Features work together. The system uses machine learning algorithms to analyse URL characteristics and classify URLs as phishing or legitimate. The input URL is processed to extract a 21-dimensional hybrid feature vector containing lexical and structural features. The extracted features are passed to the trained classification models, and the prediction results are analysed using SHAP-based explainability techniques. The system then presents the predicted class, relevant risk indicators, and explanation details through a web-based dashboard. The main aim of the system is to identify potentially malicious URLs, provide understandable predictions, and support users in making informed decisions when encountering suspicious links. The complete system must be tested to evaluate classification performance, prediction time, and the clarity of the explanations.

4.1 PROPOSED SYSTEM

The proposed system, titled PhishPulse: An Interpretable and Lightweight Framework for Phishing URL Detection Using URL Features, aims to identify potentially malicious URLs using lexical and structural characteristics. The system combines machine learning-based classification with explainability techniques and dashboard-based presentation.

The system accepts a URL as input and validates it before processing. The URL is analysed to extract a 21-dimensional hybrid feature vector containing characteristics such as URL length, suspicious tokens, special characters, character entropy, and domain structure. These features are converted into a numerical representation suitable for machine learning classification.

The proposed system uses Random Forest, Decision Tree, and Logistic Regression algorithms to classify URLs as phishing or legitimate. The models are trained using a labelled dataset containing phishing and legitimate URLs. Their predictions are compared during evaluation to assess classification performance and identify the most suitable model for the application.

The proposed system also incorporates SHAP-based explainability to identify the features influencing the classification results. These explanations help indicate why a URL may have been classified as potentially phishing or legitimate. The system presents the classification outcome and relevant risk indicators through a user-friendly dashboard.

Finally, the system provides the prediction results to the user in an understandable format. The proposed solution focuses on URL-based analysis and does not require webpage fetching or HTML extraction during URL analysis. It is designed as a software-based application using available datasets and machine learning tools, without requiring the development of custom hardware.

4.2 ADVANTAGES OF THE PROPOSED SYSTEM

The proposed system is expected to provide several advantages over approaches that rely only on basic URL checks or classification without explanations.

First, the use of machine learning algorithms supports the automatic classification of phishing and legitimate URLs, reducing the need for users to inspect every URL manually. Actual detection performance will depend on the quality of the dataset, selected features, model configuration, and experimental results.

Second, the use of lexical and structural URL features enables the system to examine suspicious patterns, including URL length, special characters, suspicious tokens, character entropy, and domain structure. Combining these characteristics provides multiple types of information for classification.

Third, the system uses a 21-dimensional hybrid feature vector to represent relevant URL characteristics in a structured numerical format. This provides a consistent input representation for training and evaluating the selected machine learning algorithms.

Fourth, the system analyses URLs directly without fetching webpage content or extracting HTML. This reduces dependence on external webpage requests and allows the detection process to focus on the information available in the URL itself.

Fifth, the use of Random Forest, Decision Tree, and Logistic Regression allows the performance of different machine learning approaches to be compared. This helps identify differences in classification effectiveness and prediction time.

Sixth, SHAP-based explainability helps identify the features that influence model predictions. Instead of presenting only a classification result, the system can display relevant feature contributions to improve the transparency of its decisions.

These advantages describe the intended benefits of the proposed design. Their effectiveness must be established through implementation, testing, and experimental evaluation.

4.3 SYSTEM ARCHITECTURE

The proposed system consists of several connected software modules. Each module performs a specific task, and the output of one module becomes the input to the next module.

The first module receives a URL from the user through the dashboard. The input URL is validated and passed to the URL preprocessing module, which prepares it for feature extraction. The hybrid feature extraction module then identifies lexical and structural characteristics and constructs a 21-dimensional feature vector.

The extracted feature vector is passed to the machine learning classification module. The trained Random Forest, Decision Tree, and Logistic Regression models process the features and generate their respective predictions. The selected model's output is used to determine the predicted class and, where supported, the associated prediction probability.

The next module performs explainability analysis using SHAP. It examines the contribution of relevant features to the model's prediction and generates information that can be used to explain the classification result. The explanation module must use an explainer compatible with the selected model.

The dashboard module receives the classification result and explanation information. It displays whether the URL is predicted to be phishing or legitimate, along with relevant risk indicators and feature contributions. The application can also record evaluation results for comparing model performance during testing.

The main components of the proposed architecture are:

- URL Validation and Preprocessing
- Hybrid Feature Extraction
- 21-Dimensional Feature Vector Construction
- Machine Learning Classification
- SHAP-Based Explainability
- Risk Indicator Generation
- Dashboard and Result Presentation

These modules form the proposed software architecture. The final implementation may be adjusted according to dataset availability, model performance, the selected backend framework, and the requirements of the dashboard. The system's predictions should be treated as risk assessments rather than guarantees that a URL is safe or malicious.

4.4 MODULE DESCRIPTION

The PhishPulse system consists of different modules that work together to detect potentially malicious URLs and provide understandable classification results. The URL input module accepts URLs from users, while the preprocessing module validates and prepares them for analysis. The hybrid feature extraction module identifies lexical and structural characteristics and constructs a 21-dimensional feature vector. The machine learning module uses Random Forest, Decision Tree, and Logistic Regression to classify URLs as phishing or legitimate. The explainability module uses SHAP to identify the features influencing predictions, and the dashboard module displays the classification results and relevant risk indicators. Together, these modules aim to provide lightweight, interpretable, and user-friendly phishing URL detection.

4.4.1 URL Input and Preprocessing

The URL input and preprocessing is the initial stage of the proposed system. Its purpose is to receive URLs and prepare them for feature extraction and classification.

The system accepts a URL entered by the user through the dashboard. The input is checked for missing values, invalid formats, and unsuitable entries. Basic URL parsing is performed to identify components such as the scheme, hostname, path, query string, and other available URL elements.

The preprocessing stage ensures that the input is represented consistently before feature extraction. Duplicate records and incorrect labels in the training dataset are also handled during dataset preparation. Training, validation, and testing data are kept separate to reduce the risk of data leakage.

The system analyses the URL string directly without fetching webpage content or extracting HTML. The output of this stage is a validated URL ready for the hybrid feature extraction module. Input validation rules will be finalised during implementation

4.4.2 Hybrid URL Feature Extraction

The hybrid feature extraction module identifies relevant characteristics of the input URL and converts them into a numerical representation for machine learning classification.

The proposed system extracts a 21-dimensional hybrid feature vector using lexical and structural URL characteristics. Lexical features examine the text and character patterns of the URL, while structural features examine the organisation of its components.

The feature set includes characteristics such as URL length, suspicious tokens, special characters, character entropy, and domain structure. These features provide information that the machine learning models can use to distinguish patterns associated with phishing and legitimate URLs.

The extracted values are arranged in a consistent order to form the feature vector. The same feature definitions, ordering, and preprocessing rules must be applied during training and prediction to ensure that the input matches the format expected by the trained models.

The output of this module is the 21-dimensional numerical feature vector, which is passed to the machine learning classification module. The usefulness of the selected features will be assessed through experimental evaluation.

4.4.3 Machine Learning Classification

The machine learning classification module is responsible for predicting whether an input URL is potentially phishing or legitimate. The proposed system uses Random Forest, Decision Tree, and Logistic Regression algorithms for this purpose.

The models are trained using a labelled dataset containing phishing and legitimate URLs. During training, the models learn relationships between the extracted URL features and their corresponding class labels. The trained models are then used to predict the class of previously unseen URLs.

When a user submits a URL, its 21-dimensional feature vector is passed to the selected trained model. The model produces a classification result indicating phishing or legitimate. Where supported, a prediction probability can also be generated and displayed.

Each algorithm has different characteristics. Random Forest combines predictions from multiple decision trees, Decision Tree uses a tree-based sequence of decision rules, and Logistic Regression estimates class probabilities using a learned relationship between input features and the target class.

4.4.4 SHAP-Based Explainability

The explainability module helps users understand which URL characteristics influenced a model's classification. It uses SHAP (SHapley Additive exPlanations) to estimate the contribution of individual features to a prediction.

After a URL has been classified, the explainability module analyses the relevant feature contributions. For tree-based models such as Random Forest and Decision Tree, SHAP TreeExplainer may be used where supported. For Logistic Regression, a suitable explainer compatible with the model and input representation can be selected.

The resulting explanations identify features that contribute positively or negatively to the model's output. For example, suspicious tokens or unusual domain structures may contribute to a phishing prediction if the trained model has learned an association with those features. The actual contributions depend on the input URL and the trained model.

The explanation module provides information to the dashboard, where relevant feature contributions can be displayed using a feature breakdown or similar visualisation. These explanations describe the model's behaviour; they do not independently establish whether a URL is malicious.

This module aims to improve transparency by providing more information than a classification label alone. The correctness and usefulness of the explanations will be assessed during testing.

4.4.5 Risk Indicator Generation

The risk indicator module converts the extracted URL characteristics and classification results into information that users can understand more easily.

The system identifies relevant URL properties, such as unusual URL length, suspicious tokens, special characters, character entropy, and domain structure. These indicators are presented alongside the model's prediction to help users understand potential warning signs.

The risk indicators are generated from the available URL features and model explanations. Their presentation should remain consistent with the evidence provided by the analysis. A suspicious feature should not automatically be treated as proof of phishing, because legitimate URLs can also contain unusual patterns.

The module may display the predicted class, selected risk indicators, and relevant SHAP feature contributions. If a probability or risk score is displayed, its meaning should be clearly explained and should not be presented as a guaranteed measure of maliciousness unless it has been appropriately calibrated and validated.

4.4.6 Dashboard and Result Presentation

The dashboard module provides the user interface for interacting with the PhishPulse system. It allows users to submit URLs and view the classification results and corresponding explanations.

When a URL is submitted, the frontend sends the input to the backend for validation, feature extraction, and classification. The backend returns the prediction and the available explanation data to the frontend.

The dashboard displays whether the URL has been classified as phishing or legitimate. It can also present relevant risk indicators, feature contributions, and prediction information in a clear format. Appropriate visual elements, such as charts or feature breakdowns, may be used to improve readability.

The interface should distinguish the predicted class from the explanation details and communicate that the result is a machine learning assessment. A legitimate prediction does not guarantee that a URL is safe, and a phishing prediction should be interpreted in the context of the available evidence.

The dashboard is intended to make the detection process accessible to users who may not have technical knowledge of machine learning. Its usability and correct display of results will be checked during application testing.

4.4.7 Model Evaluation and Performance Analysis

The model evaluation module measures the performance of the selected algorithms and helps identify limitations in phishing URL detection.

The Random Forest, Decision Tree, and Logistic Regression models are evaluated using a separate test dataset that is not used for model training. The evaluation considers the number of correct and incorrect phishing and legitimate predictions.

The following metrics are used:

- Accuracy: Measures the proportion of all predictions that are correct.
- Precision: Measures the proportion of URLs predicted as phishing that are actually labelled phishing.
- Recall: Measures the proportion of phishing URLs identified by the model.
- F1-score: Combines precision and recall into a single measure.

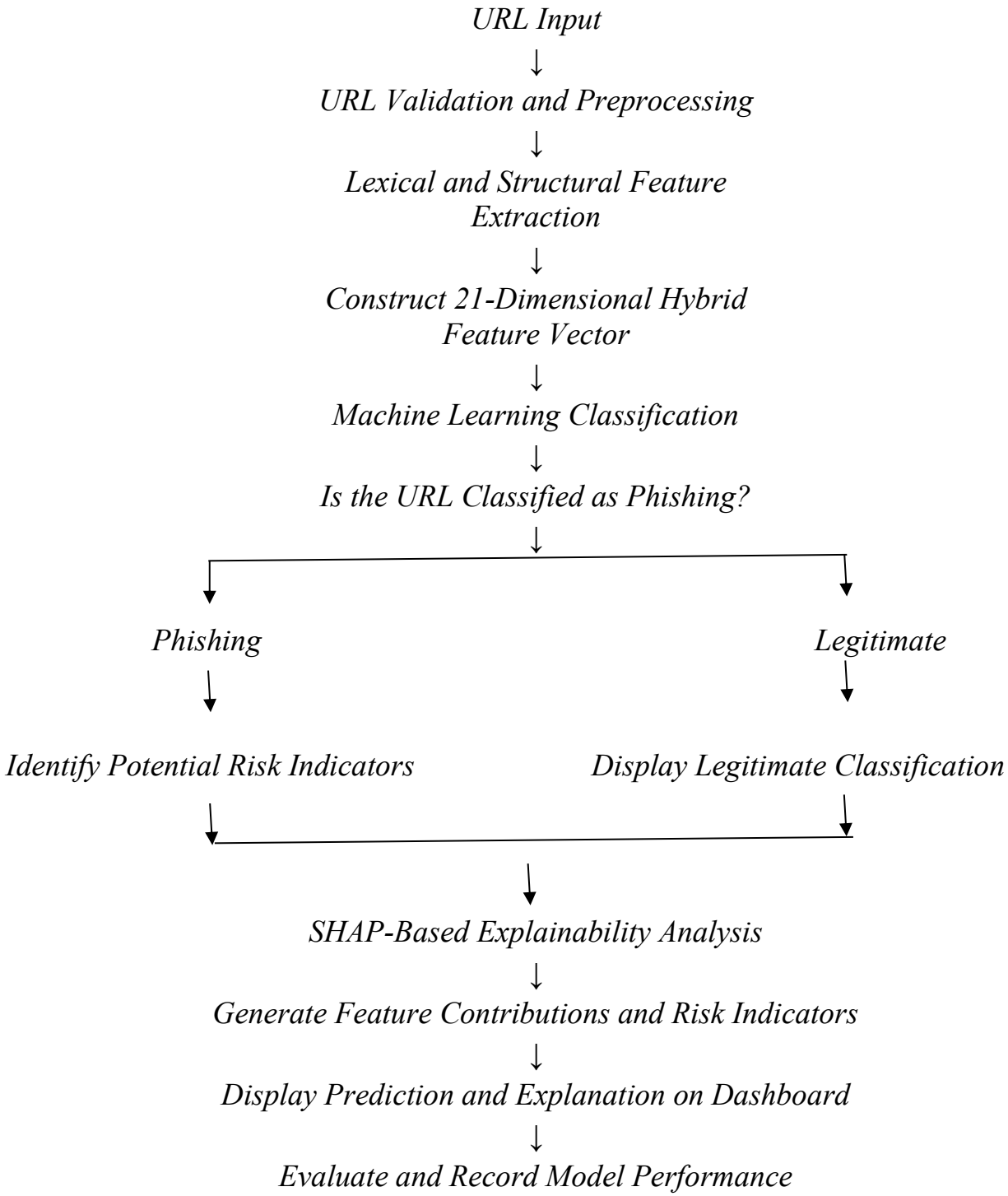
- Prediction time: Measures the time required to process an input and produce a prediction under the defined testing conditions.

The evaluation should also examine false positives, where legitimate URLs are classified as phishing, and false negatives, where phishing URLs are classified as legitimate. These errors are important because they affect user trust and the effectiveness of the detection system.

4.5 SYSTEM FLOW DIAGRAM

The proposed PhishPulse system follows a sequence of stages, beginning with URL input and ending with phishing classification, explainability analysis, and result presentation through the dashboard.

Proposed system flow:



The system validates the input URL and extracts lexical and structural characteristics without fetching webpage content or performing HTML scraping. The resulting 21-dimensional feature vector is passed to the trained machine learning model to classify the URL as phishing or legitimate.

Following classification, SHAP-based explainability identifies the features that influenced the prediction. The dashboard displays the classification result, relevant risk indicators, and feature contributions to help users understand the model's decision.

If the URL is classified as phishing, the system highlights potential warning signs. If it is classified as legitimate, the result is displayed accordingly; however, this classification does not guarantee that the URL is safe.

The extracted features, trained models, and explanation process must remain consistent between training and prediction. The system should also be tested using previously unseen URLs to evaluate classification performance and identify false-positive and false-negative predictions.

4.6 DATASET DESCRIPTION

The dataset is an important component in developing and evaluating the PhishPulse phishing URL detection system. It should contain labelled examples of phishing and legitimate URLs with sufficient variation in URL patterns, domain structures, suspicious tokens, and character combinations.

The proposed system may use an existing publicly available phishing URL dataset, provided that its contents, label quality, licensing conditions, and suitability for the project are verified. Additional URL samples may be considered if they can be collected and used appropriately.

The dataset should ideally include different types of URLs, including URLs with varying lengths, special characters, subdomain structures, suspicious keywords, and character patterns. Both phishing and legitimate examples are required to help the machine learning models learn differences between the two classes. Diverse samples can also help assess the system's ability to handle previously unseen URLs.

For supervised machine learning, each URL must have a suitable class label indicating whether it is phishing or legitimate. The URLs are processed to extract a 21-dimensional hybrid feature vector containing lexical and structural characteristics.

The dataset should be divided into training, validation, and testing sets. The training set is used to train the models, the validation set supports model and threshold selection, and the testing set is reserved for final evaluation. Duplicate URLs and closely related samples should be handled carefully to reduce data leakage and avoid misleading performance results.

Detection performance can be evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. Prediction time may also be measured to assess the processing efficiency of the system. The evaluation should consider false-positive predictions, where legitimate URLs are classified as phishing, and false-negative predictions, where phishing URLs are classified as legitimate.

The dataset supports URL classification and feature analysis, but it does not necessarily provide explanations for individual predictions. SHAP-based explainability will be applied to the trained models to examine the contribution of extracted features to their predictions. This helps connect the dataset and classification process with the interpretability objectives of PhishPulse.

The exact dataset name, number of URLs, label distribution, feature definitions, train-validation-test split, and final evaluation results will be reported after the dataset has been selected and examined. No dataset size or performance result is assumed at this stage..

4.7 SYSTEM MAINTENANCE

System maintenance is required to keep the proposed PhishPulse software reliable as new phishing techniques emerge and URL patterns change.

The dataset should be checked periodically for missing values, invalid URLs, duplicate records, and incorrect class labels. Maintaining data quality helps reduce errors during model training and evaluation.

The phishing URL detection models may need to be reassessed if their performance decreases on new or previously unseen URL patterns. Any retraining or model replacement should use an appropriate labelled dataset and be followed by testing to determine whether detection performance has improved.

The URL feature extraction module should also be reviewed to ensure that the 21-dimensional hybrid feature vector is generated consistently. Changes to feature definitions, ordering, or preprocessing rules must be reflected in both model training and prediction to maintain compatibility with the trained models.

The explainability module and dashboard should be checked to ensure that SHAP explanations, classification results, and risk indicators are displayed correctly. Model updates should also be followed by appropriate checks to confirm that the explanations remain compatible with the selected model.

The system should record relevant errors so that problems in URL validation, feature extraction, model prediction, explanation generation, or dashboard communication can be investigated.

Suitable backups and access controls should be considered if the system stores datasets, trained models, or evaluation records. User-submitted URLs and other potentially sensitive information should be handled appropriately and retained only when necessary.

System maintenance procedures will be defined more precisely after the software architecture and implementation choices are finalized. The effectiveness of these procedures will depend on the completed system and its evaluation.

CHAPTER 5

SYSTEM IMPLEMENTATION

The system implementation involves developing PhishPulse: An Interpretable and Lightweight Framework for Phishing URL Detection Using URL Features. The system analyses URLs using a 21-dimensional hybrid feature vector containing lexical and structural characteristics. Random Forest, Decision Tree, and Logistic Regression algorithms are used to classify URLs as phishing or legitimate. A Python-based backend handles URL preprocessing, feature extraction, model inference, and explainability, while a web-based frontend provides an interface for entering URLs and viewing prediction results. SHAP is used to explain the features influencing the model's predictions, and the dashboard displays classification results and relevant risk indicators. The system analyses URLs directly without fetching webpage content or extracting HTML. Its implementation is evaluated using classification metrics such as accuracy, precision, recall, F1-score, and ROC-AUC, along with prediction time and the clarity of the explanations.

5.1 SOFTWARE SETUP AND ENVIRONMENT

The software environment is an important part of developing the PhishPulse system. It provides the tools and libraries required to process URLs, extract features, train machine learning models, generate explanations, and display classification results.

Python is proposed as the primary programming language because it supports data processing, machine learning, URL analysis, and backend development. The development environment will be selected according to the available computing resources and the requirements of the selected algorithms.

Pandas and NumPy may be used for dataset handling, data cleaning, and numerical operations. Scikit-learn will support the implementation and evaluation of Random Forest, Decision Tree, and Logistic Regression models. SHAP will be used to explain the features influencing the model predictions.

A suitable backend framework, such as Flask or FastAPI, may be used to connect the frontend with the feature extraction and classification modules. HTML, CSS, and JavaScript may be used to develop the user interface, while Matplotlib or Plotly can support the visualisation of model performance and feature contributions.

The proposed software environment includes the following components:

- Python: Application development, URL feature extraction, and model processing.
- Pandas and NumPy: Dataset processing and numerical feature handling.
- Scikit-learn: Training and evaluation of the selected machine learning algorithms.
- SHAP: Explainability analysis of model predictions.
- Flask or FastAPI: Backend development and communication between application modules.
- HTML, CSS, and JavaScript: Frontend development and user interaction.
- Matplotlib or Plotly: Visualisation of performance metrics and feature contributions.
- CSV: Storage and management of labelled URL datasets and evaluation results.
- JSON: Communication of input URLs, classification results, and explanation data between the frontend and backend.

These are proposed tools and technologies. The final versions, dependencies, and configurations will be documented after the development environment is established. The project does not require the design or manufacture of custom hardware.

5.2 IMPLEMENTATION OF PHISHING URL DETECTION

The phishing URL detection module identifies potentially malicious URLs by analysing their lexical and structural characteristics. It is one of the main components of PhishPulse because the explainability and dashboard modules depend on its classification results.

The implementation begins by selecting a suitable labelled dataset containing phishing and legitimate URLs. The dataset is examined for label quality, duplicate records, missing values, and suitability for the project. It is then divided into training, validation, and testing sets to support model development and reliable performance evaluation.

The URL feature extraction process identifies relevant characteristics and constructs a 21-dimensional hybrid feature vector. These features include URL length, suspicious tokens, special characters, character entropy, and domain structure.

The extracted values are converted into a numerical representation suitable for the selected machine learning algorithms.

Random Forest, Decision Tree, and Logistic Regression models are then trained using the prepared training dataset. The validation set may be used to select suitable model settings and classification thresholds.

During operation, the user enters a URL through the dashboard. The system validates the input, extracts the required features, and passes the resulting feature vector to the selected trained model. The model produces a classification indicating whether the URL is phishing or legitimate. Where supported, a prediction probability may also be displayed.

The output of this module is the classification result, which is passed to the explainability module for further analysis. The detection process focuses on the URL string without fetching webpage content or extracting HTML.

The module will be evaluated using a separate test dataset to measure classification performance. Accuracy, precision, recall, F1-score, ROC-AUC, and prediction time will be reported only after the experiments have been completed.

5.3 IMPLEMENTATION OF HYBRID URL FEATURE EXTRACTION

The hybrid URL feature extraction module converts the characteristics of an input URL into a numerical representation that can be processed by the machine learning models.

The implementation begins by validating and parsing the URL to identify available components, such as the hostname, path, query string, and special characters. The system then calculates the selected lexical and structural features according to predefined feature extraction rules.

Lexical features examine the textual characteristics of the URL, including its length, suspicious tokens, and special-character patterns. Structural features examine the organisation of the URL, including domain structure and subdomain patterns. Character entropy may also be calculated to represent the variation in characters within the URL.

The extracted features are combined to construct the proposed 21-dimensional hybrid feature vector. Each dimension represents a predefined URL characteristic, and the features must be arranged in a consistent order.

The same feature extraction rules must be applied during dataset preparation, model training, validation, testing, and real-time prediction. Any required scaling or other learned preprocessing must be fitted using the training data and then applied consistently to the remaining datasets.

The implementation will be tested using different types of phishing and legitimate URLs to identify errors in URL parsing, feature calculation, and feature vector construction. The exact definitions of all 21 features will be documented in the final implementation.

5.4 IMPLEMENTATION OF EXPLAINABILITY AND DASHBOARD MODULES

The The explainability and dashboard modules help users understand the classification results produced by the PhishPulse system.

The explainability module uses SHAP (SHapley Additive exPlanations) to identify the features that influence a model's prediction. After a URL has been classified, the system analyses the contributions of the extracted features to the model output.

For Random Forest and Decision Tree, SHAP TreeExplainer may be used where supported. For Logistic Regression, a suitable explainer compatible with the model and input representation will be selected. The explanation method must be configured according to the selected model and its output format.

The explanation results may identify features that contribute positively or negatively to the predicted class. For example, suspicious tokens, unusual character patterns, or complex domain structures may influence the prediction if the trained model has learned associations between these features and phishing URLs.

The dashboard provides the user interface for submitting URLs and viewing the results. The frontend sends the URL to the backend, which performs validation, feature extraction, classification, and explainability analysis. The backend then returns the available results to the frontend.

The dashboard may display the following information:

- Input URL submitted by the user.
- Predicted class: phishing or legitimate.
- Prediction probability, where supported.
- Relevant URL risk indicators.
- SHAP-based feature contributions.
- Model evaluation results, where available.

The interface should present the results in a clear and understandable format. A phishing prediction indicates potential risk, while a legitimate prediction does not guarantee that the URL is safe. Similarly, SHAP explanations describe the model's behaviour and do not independently prove whether a URL is malicious.

5.5 TESTING METHODOLOGY

Testing will be conducted to evaluate the correctness and reliability of the individual modules and the integrated PhishPulse system. The objective is to identify errors, measure performance, and determine whether the proposed system meets its intended requirements.

The testing process will include the following areas:

URL validation testing: The input module will be tested using valid URLs, malformed URLs, empty inputs, and URLs containing unusual characters. The tests will examine whether the system handles acceptable and invalid inputs appropriately.

Feature extraction testing: The feature extraction module will be tested to verify that the 21-dimensional hybrid feature vector is generated correctly. The tests will check feature values, feature ordering, and consistency between training and prediction.

Machine learning model testing: Random Forest, Decision Tree, and Logistic Regression will be evaluated using a separate test dataset. The predictions will be compared with the reference labels to identify correct classifications, false positives, and false negatives.

Explainability testing: The SHAP module will be checked to ensure that the explanations correspond to the selected model and input features. The tests will examine whether the feature contributions are generated and presented correctly.

Dashboard testing: The frontend will be tested to confirm that URLs can be submitted and that classification results, risk indicators, and explanation details are displayed correctly. Error messages and invalid inputs will also be examined.

Integration testing: The modules will be tested together to verify the flow of information from URL input and feature extraction through model classification, explainability analysis, and dashboard presentation.

Performance testing: The system will be tested to measure classification metrics and prediction time under the defined experimental conditions. The comparison will help identify differences between the selected algorithms.

Test cases, expected outcomes, actual outcomes, and errors will be documented. The final report will distinguish successful tests from failed tests and unresolved issues. All performance results will be based on actual experiments.

5.6 EVALUATION METRICS

Appropriate **5.6 EVALUATION METRICS**

Appropriate evaluation metrics are used to assess the classification performance and processing efficiency of the PhishPulse system. These metrics help determine how effectively the models identify phishing URLs and distinguish them from legitimate URLs.

Accuracy

Accuracy measures the proportion of all predictions that are correct.

Accuracy = (TP + TN) / (TP + TN + FP + FN)

Here, TP represents True Positives, TN represents True Negatives, FP represents False Positives, and FN represents False Negatives.

Precision

Precision measures the proportion of URLs predicted as phishing that are actually labelled as phishing.

Precision = TP / (TP + FP)

Recall

Recall measures the proportion of actual phishing URLs that are correctly identified by the model.

Recall = TP / (TP + FN)

F1-Score

The F1-score combines precision and recall into a single measure.

F1-Score = 2 × (Precision × Recall) / (Precision + Recall)

ROC-AUC

The Area Under the Receiver Operating Characteristic Curve (ROC-AUC) measures how well the model distinguishes between phishing and legitimate URLs across different classification thresholds. A higher ROC-AUC generally indicates better discrimination between the two classes.

Prediction Time

Prediction time measures the time required to process an input URL and produce a classification result. It is measured under consistent hardware and software conditions to compare the processing efficiency of the selected models.

Explainability Assessment

The explainability component is examined to determine whether SHAP explanations are generated correctly and whether the identified feature contributions are presented clearly.

Dashboard Functionality

The dashboard is tested to confirm that classification results, risk indicators, and explanation details match the backend output. Invalid input handling and error reporting are also considered.

The final evaluation uses metrics that can be measured reliably with the available dataset and testing environment. Numerical results will be included only after the experiments have been performed.

5.7 LIMITATIONS OF THE SYSTEM

The proposed PhishPulse system may face several limitations that can affect its classification performance and practical usefulness.

The accuracy of phishing URL detection depends on the quality, diversity, and representativeness of the training dataset. If the dataset contains incorrect labels, duplicated samples, or limited URL patterns, the trained models may not perform reliably on previously unseen URLs.

URL-based analysis is another limitation. Some phishing URLs may appear similar to legitimate URLs, while legitimate websites may contain unusual characters, long paths, or complex subdomains. Consequently, lexical and structural features alone may not identify every phishing URL or correctly classify every legitimate URL.

The performance of Random Forest, Decision Tree, and Logistic Regression may vary depending on the selected features, training data, model configuration, and classification threshold.

Explainability also has limitations. SHAP identifies how features contribute to a model's prediction, but these contributions do not establish that a URL is definitively malicious or safe

The processing speed of the system may vary according to the selected model, feature extraction procedure, hardware, and software configuration. Prediction time must therefore be measured during implementation rather than assumed from the algorithms used.

Finally, the system does not fetch webpage content or extract HTML during URL analysis. Although this keeps the detection process focused on URL characteristics, it also means the system does not examine webpage content or other external indicators that might help identify some phishing attacks.

These limitations will be documented and considered when interpreting the final results. The effectiveness of PhishPulse will be determined through systematic testing and experimental evaluation rather than assumed performance.

CHAPTER 6

RESULT AND DISCUSSION

6.1 OBSERVATIONS

This section presents the observations obtained from the implementation and evaluation of the proposed PhishPulse phishing URL detection system. The performance of the Random Forest, Decision Tree, and Logistic Regression models is analysed using accuracy, precision, recall, F1-score, ROC-AUC, and prediction time. The observations help identify the strengths and limitations of the system in distinguishing phishing URLs from legitimate URLs.

The evaluation results will be obtained by testing the trained models on a separate test dataset containing phishing and legitimate URLs that were not used during training. The performance of each model will be measured using the selected classification metrics. Prediction time will also be recorded to assess the processing efficiency of the system. SHAP-based explanations will be examined to identify the URL features influencing the classification results.

The observations will help determine how effectively the system identifies phishing URLs and how consistently it classifies legitimate URLs. The results will also be used to compare the three machine learning algorithms and examine the contribution of the 21-dimensional hybrid feature vector to the detection process. Any differences between training, validation, and test performance will be considered when assessing the model's ability to generalise to previously unseen URLs.

The system's limitations will also be examined by reviewing false-positive predictions, where legitimate URLs are classified as phishing, and false-negative predictions, where phishing URLs are classified as legitimate. The explainability results and dashboard output will be checked to ensure that the displayed information corresponds to the model predictions.

Actual numerical results will be added after the implementation and experiments have been completed. No accuracy values, performance measurements, or experimental findings are assumed at this stage.

6.2 DISCUSSION AND FINDINGS

The evaluation of PhishPulse is intended to determine whether combining lexical and structural URL features with machine learning classification provides a useful approach to phishing URL detection. Random Forest, Decision Tree, and Logistic Regression will be compared to identify differences in their classification performance and processing efficiency.

Accuracy indicates the proportion of correctly classified URLs, while precision measures how reliably the models identify URLs predicted as phishing. Recall indicates the proportion of actual phishing URLs successfully detected. The F1-score provides a combined measure of precision and recall, while ROC-AUC evaluates the models' ability to distinguish between phishing and legitimate URLs across different classification thresholds. Prediction time will be used to compare the processing efficiency of the selected algorithms.

The system's use of a 21-dimensional hybrid feature vector enables the models to analyse lexical and structural characteristics, including URL length, suspicious tokens, special characters, character entropy, and domain structure. However, URL-based analysis has limitations because some phishing URLs may resemble legitimate URLs, while legitimate URLs may contain suspicious-looking patterns. The system's effectiveness must therefore be established through testing on suitable, previously unseen data.

SHAP-based explainability is intended to improve the transparency of the classification process by identifying features that influence model predictions. The dashboard presents the predicted class, relevant risk indicators, and available feature contributions to help users understand the results. However, SHAP explanations describe the behaviour of the trained model and do not independently prove that a URL is malicious or safe.

Overall, the project aims to establish a framework for lightweight and interpretable phishing URL detection. Further improvements may focus on reducing false-positive and false-negative predictions, evaluating generalisation to new URL patterns, and improving the clarity of the explanations presented to users. The final findings will be based on the measured experimental results, and comparisons with related research will account for differences in datasets, evaluation metrics, model configurations, and testing conditions.

CHAPTER 7

CONCLUSION

7.1 SUMMARY

Phishing attacks are a major cybersecurity concern because they use deceptive URLs to trick users into visiting fraudulent websites and revealing sensitive information. Identifying potentially malicious URLs and providing understandable classification results can improve user awareness and support safer online activities. This project proposes PhishPulse An Interpretable and Lightweight Framework for Phishing URL Detection Using URL Features, which integrates URL feature extraction, machine learning classification, explainability analysis, and dashboard-based result presentation.

The system uses lexical and structural URL characteristics to construct a 21-dimensional hybrid feature vector. These features include URL length, suspicious tokens, special characters, character entropy, and domain structure. The extracted features are processed by Random Forest, Decision Tree, and Logistic Regression algorithms to classify URLs as phishing or legitimate

The proposed explainability mechanism uses SHAP to identify the features influencing the model's predictions. These explanations are intended to help users understand why a URL may have been classified as potentially phishing or legitimate. The dashboard presents the classification results, relevant risk indicators, and available feature contributions in an understandable format.

The model evaluation process considers accuracy, precision, recall, F1-score, ROC-AUC, and prediction time. These metrics will be used to compare the selected machine learning algorithms, identify classification errors, and assess the processing efficiency of the system. Testing will also examine whether the SHAP explanations and dashboard outputs.

At the present stage, the project has established its proposed design and implementation approach. The feature extraction module, machine learning models, explainability component, and dashboard require implementation and experimental evaluation. Further testing using suitable datasets and previously unseen URLs is necessary to determine whether PhishPulse meets its objectives. The final effectiveness of the system will depend on the measured classification performance, reliability of the explanations, and consistency of the complete application.

LIST OF PUBLICATIONS

1. Conference / Journal Publication: Not yet published. Status: Research and planning stage; manuscript preparation is in progress.

APPENDIX 1

SYSTEM REQUIREMENTS AND TECHNICAL DETAILS

1. **Hardware Requirements**
   - Computer or laptop for system development and execution.
   - Sufficient RAM and storage for dataset processing and model training.
   - Processor capable of executing machine learning models.
   - Internet connection for downloading datasets and accessing the web application during development.
2. **Software Requirements**
   - Python for application development and machine learning.
   - Scikit-learn for Random Forest, Decision Tree, and Logistic Regression.
   - Pandas and NumPy for dataset processing and feature extraction.
   - SHAP for explainability analysis.
   - Flask or FastAPI for backend development.
   - HTML, CSS, and JavaScript for frontend development.
   - Matplotlib or Plotly for visualisation.
3. **Main System Components**
   - URL Input and Validation
   - URL Preprocessing
   - Hybrid URL Feature Extraction
   - 21-Dimensional Feature Vector Construction
   - Machine Learning Classification
   - SHAP-Based Explainability
   - Risk Indicator Generation
   - Dashboard and Result Presentation
   - Model Performance Evaluation
4. **Evaluation Metrics**
   - Accuracy
   - Precision
   - Recall
   - F1-score
   - ROC-AUC
   - Prediction Time

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